Practice Assignment 6 - Not Graded

This assignment will not be graded and is only for practice.
Multiple Choice Questions (MCQ):

1 point
Which of the following statements are correct?

☐ There exists a linear transformation $T: R^2 \rightarrow R^2$ T : R2 → R2 such that $T(2,3) = (1,2)$T(2, 3) = (1, 2), $T(1,-1) = (1,-1)$T(1, −1) = (1, −1), and $T(4,1) = (1,0)$T(4, 1) = (1, 0).

☐ If there is a linear transformation $T: R^2 \rightarrow R^2$ T : R2 → R2 such that $T(2,3) = (1,2),$ and $T(1,-1) = (1,-1)$T(2, 3) = (1, 2), and T(1, −1) = (1, −1), then $T(x,y) = \frac{1}{5}(4x-y,-x+4y)$T(x, y) = 51(4x − y, −x + 4y).

☐ Let $\beta = \{v_1, v_2, v_3\}$β = {v1, v2, v3} and $\gamma = \{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$γ = {2v1 + v3, v2 − v3, v3 − v1} be bases of a vector space VV. Consider a linear transformation $T: V \rightarrow V$T : V → V such that $T(v_1) = 2v_3 + v_1, T(v_2) = 2v_1,$ and $T(v_3) = 2v_2$ T(v1) = 2v3 + v1, T(v2) = 2v1, and T(v3) = 2v2. The matrix representation of TT, with respect to ββ and γγ for the domain and codomain respectively, is $\begin{bmatrix} 1 & \frac{2}{3} & \frac{2}{3} \\ 0 & 0 & 2 \\ 1 & -\frac{24}{33} & \end{bmatrix}$ $\begin{bmatrix} & & \\ 1 & 0 & 1 \\ 3 & 2 & 0 \\ -3 & 2 & 3 \\ 4 & & \end{bmatrix}$

☐ Let $\{v_1, v_2, v_3\}$ {v1, v2, v3} be a basis of a vector space VV and $\{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$ {2v1 + v3, v2 − v3, v3 − v1} be a basis of vector space WW. Consider a linear transformation $T: V \rightarrow W$ T : V → W such that $T(v_1) = 2v_3 + v_1, T(v_2) = 2v_1,$ and $T(v_3) = 2v_2$ T(v1) = 2v3 + v1, T(v2) = 2v1, and T(v3) = 2v2. Then, the rank of TT is 3.

No, the answer is incorrect.
Score: 0
Accepted Answers:

If there is a linear transformation $T: R^2 \rightarrow R^2$ T : R2 → R2 such that $T(2,3) = (1,2),$ and $T(1,-1) = (1,-1)$T(2, 3) = (1, 2), and T(1, −1) = (1, −1), then $T(x,y) = \frac{1}{5}(4x-y,-x+4y)$T(x, y) = 51(4x − y, −x + 4y).
Let $\beta = \{v_1, v_2, v_3\}$β = {v1, v2, v3} and $\gamma = \{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$γ = {2v1 + v3, v2 − v3, v3 − v1} be bases of a vector space VV. Consider a linear transformation $T: V \rightarrow V$T : V → V such that $T(v_1) = 2v_3 + v_1, T(v_2) = 2v_1,$ and $T(v_3) = 2v_2$ T(v1) = 2v3 + v1, T(v2) = 2v1, and T(v3) = 2v2. The matrix representation of TT, with respect to ββ and γγ for the domain and codomain respectively, is $\begin{bmatrix} 1 & \frac{2}{3} & \frac{2}{3} \\ 0 & 0 & 2 \\ 1 & -\frac{24}{33} & \end{bmatrix}$ $\begin{bmatrix} & & \\ 1 & 0 & 1 \\ 3 & 2 & 0 \\ -3 & 2 & 3 \\ 4 & & \end{bmatrix}$
Let $\{v_1, v_2, v_3\}$ {v1, v2, v3} be a basis of a vector space VV and $\{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$ {2v1 + v3, v2 − v3, v3 − v1} be a basis of vector space WW. Consider a linear transformation $T: V \rightarrow W$ T : V → W such that $T(v_1) = 2v_3 + v_1, T(v_2) = 2v_1,$ and $T(v_3) = 2v_2$ T(v1) = 2v3 + v1, T(v2) = 2v1, and T(v3) = 2v2. Then, the rank of TT is 3.

1 point

Consider the following set $S = \{(x, y) \mid x + y = 1, x, y \in R\}$S = {(x, y) | x + y = 1, x, y ∈ R}. In column A the matrix representation of some linear transformations are given with respect to the standard ordered bases. Match the entries in column A with the set $T(S)$T (S) given in column B and the geometric representations of SS and $T(S)$T (S) in column C.

- [ ] i → → b → → 2.
- [ ] ii → → c → → 1.
- [ ] ii → → c → → 2.
- [ ] iii → → a → → 3.
- [ ] i → → b → → 3.

iii → → a → → 1.

No, the answer is incorrect.

Score: 0

Accepted Answers:

ii → → c → → 2.
i → → b → → 3.
iii → → a → → 1.

Multiple Select Questions (MSQ):

1 point

Consider the following two groups. In Group 1, there are 3 functions from one vector space to another, and in Group 2, there are three properties of functions. All the vector spaces have usual addition and scalar multiplication.

Group-1

- **P:** $T: R^2 \rightarrow R^2$ **T : R2 → R2 such that** $T(x, y) = \begin{cases} (\frac{x^2}{y}, y) \text{ if } y \neq 0 \\ (x, 0) \text{ otherwise} \end{cases}$ **T (x, y) = ⌊ ı ⎨ ı ⌈(yx2, y)**

(x, 0)if y = 0otherwise

- **Q:** $T: R^2 \rightarrow R^2$ **T : R2 → R2 such that** $T(x, y) = (x, xy)$ **T (x, y) = (x, xy)**
- **R:** $T: R^3 \rightarrow R$ **T : R3 → R such that** $T(x, y, z) = \frac{x + y + z}{5}$ **T (x, y, z) = 5x+y+z**

Group-2

- 1: $T(v_1 + v_2) = T(v_1) + T(v_2)$ T (v1 + v2) = T (v1) + T (v2) for all $v_1, v_2 \in V$ v1, v2 ∈ V
- 2: $T(cv) = cT(v)$ T (cv) = cT (v) for all $v \in V$ v ∈ V and $c \in R$ c ∈ R.
- 3: T T is a linear transformation

Choose the set of correct options.

- ☐ P satisfies 1 but not 2.
- ☐ P satisfies 2 but not 1
- ☐ Q satisfies 1
- ☐ Q satisfies 3
- ☐ P satisfies 3
- ☐ R satisfies 2
- ☐ R satisfies 3
- ☐ R satisfies 1 but not 2

No, the answer is incorrect.

Score: 0

Accepted Answers:

P satisfies 2 but not 1
R satisfies 2
R satisfies 3

Numerical Answer Type (NAT):

Suppose the matrix representation of a linear transformation $T: R^3 \rightarrow R^3$ T : R3 → R3 with respect to the ordered bases $\beta = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$ β = {(1, 0, 0), (1, 1, 0), (1, 1, 1)} for the domain and $\gamma = \{(1, 1, 1), (0, 1, 1), (0, 0, 1)\}$ γ = {(1, 1, 1), (0, 1, 1), (0, 0, 1)} for the range, is $I_{3\times 3}$ I3×3, i.e., the

identity matrix of order 3. Let AA be the matrix representation of the linear transformation TT with respect to standard ordered basis of R^3R3 for both domain and range. Then the determinant of AA is

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point
Let TT and SS be two linear transformations from R^3R3 to R^3R3, which are defined as follows:

$$T: R^3 \rightarrow R^3 \text{T : R3} \rightarrow \text{R3} \qquad S: R^3 \rightarrow R^3 \text{S : R3} \rightarrow \text{R3}$$

$T(x, y, z) = (x - y, y - z, z - x)$T(x, y, z) = (x − y, y − z, z − x)
$S(x, y, z) = (x + y, y + z, z + x)$S(x, y, z) = (x + y, y + z, z + x)

Let $S + T$S + T be defined to be the linear transformation as follows:

$(S + T): R^3 \rightarrow R^3$(S + T) : R3 → R3

$(S + T)(x, y, z) = S(x, y, z) + T(x, y, z)$(S + T)(x, y, z) = S(x, y, z) + T(x, y, z)

Let CC be the matrix representation of $S + T$S + T with respect to the standard ordered bases of R^3 R3. If $C = nI$C = nI, where II denotes the identity matrix of order 3, then what will be the value of n n?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point

1 point
Consider a linear transformation $T: R^3 \rightarrow R^3$T : R3 → R3 defined as $T(x, y, z) = (x + 5y, z - 3x, y + 6z)$T(x, y, z) = (x + 5y, z − 3x, y + 6z). Which of the following options are true about TT?

☐ Nullity of TT is 00.

☐ A basis for the range of TT is $\{(1, 5, 0), (0, 1, 6)\}${(1, 5, 0), (0, 1, 6)}.

☐ A basis for the range of TT is $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}${(1, 0, 0), (0, 1, 0), (0, 0, 1)}.

☐ Rank of TT is 22.

☐ The matrix representation of TT with respect to the standard ordered bases of R^3R3 is $\begin{bmatrix} 1 & -3 & 0 \\ 5 & 0 & 1 \\ 0 & 1 & 6 \end{bmatrix}$ $\begin{bmatrix} 1 & 5 & 0 \\ -3 & 0 & 1 \\ 0 & 1 & 6 \end{bmatrix}$.

☐

The matrix representation of TT with respect to the standard ordered bases of R^3 R3 is $\begin{bmatrix} 1 & 5 & 0 \\ -3 & 0 & 1 \\ 1 & -3 & 0 \\ 5 & 0 & 1 \end{bmatrix}$.

☐

Nullity of TT is 11.

☐

Rank of TT is 33.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Nullity of TT is 00.

A basis for the range of TT is $\{(1,0,0),(0,1,0),(0,0,1)\}\{(1,0,0),(0,1,0),(0,0,1)\}$.

The matrix representation of TT with respect to the standard ordered bases of R^3 R3 is $\begin{bmatrix} 1 & 5 & 0 \\ -3 & 0 & 1 \\ 1 & -3 & 0 \\ 5 & 0 & 1 \end{bmatrix}$.

Rank of TT is 33.

Let VV be a vector space with ordered basis $\alpha = \{v_1, v_2, \ldots, v_{10}\}$α = {v1, v2, … , v10}. Consider a linear transformation $T: V \rightarrow V$T : V → V such that $T(v_1) = v_1$ T(v1) = v1 and $T(v_i) = v_i - v_{i-1}$ T(vi) = vi − vi−1, where $i > 1$i > 1. Let AA be the matrix representation of the linear transformation TT, with respect to αα for both the domain and codomain. what is the trace of A A?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 10

1 point

Comprehension Type Question:

For each video on YouTube, consider the vector (Number of comments in thousands, Number of shares in thousands, Number of views in thousands) and consider the subset SS of R^3 R3 consisting of these vectors. It is known that the ranking of a YouTube video can be thought of as the restriction of a linear transformation $T: R^3 \rightarrow R$T : R3 → R to SS where R^3 R3 comes with usual addition and scalar multiplication. (assume ranking 0 (zero) is possible and in practice negative rankings can be ignored). The following data in Table M2W6P2 shows the correlation between comments, shares, and views - with YouTube rankings of a video.

Use the above information to answer questions 8, 9 and 10.

1 point

Which of the following sets of vectors in SS have the property that the YouTube video corresponding to every linear combination of that set which is in SS has ranking 00?

- [ ] {(1, 1, 0), (0, 0, 1)}{(1, 1, 0), (0, 0, 1)}
- [ ] {(6, 25, 0), (0, 0, 25)}{(6, 25, 0), (0, 0, 25)}
- [ ] {(25, 6, 0), (0, 0, 25)}{(25, 6, 0), (0, 0, 25)}
- [ ] {(12, 50, 0), (0, 0, 1)}{(12, 50, 0), (0, 0, 1)}
- [ ] {(24, 100, 0), (0, 0, 25)}{(24, 100, 0), (0, 0, 25)}
- [ ] {(6, 0, 0), (0, 25, 0), (0, 0, 1)}{(6, 0, 0), (0, 25, 0), (0, 0, 1)}
- [ ] {(25, 0, 0), (0, 6, 0), (0, 0, 1)}{(25, 0, 0), (0, 6, 0), (0, 0, 1)}

No, the answer is incorrect.

Score: 0

Accepted Answers:

{(6, 25, 0), (0, 0, 25)}{(6, 25, 0), (0, 0, 25)}

{(12, 50, 0), (0, 0, 1)}{(12, 50, 0), (0, 0, 1)}

{(24, 100, 0), (0, 0, 25)}{(24, 100, 0), (0, 0, 25)}

1 point

Which of the following sets of vectors in SS have the property that the YouTube video corresponding to every linear combination of that set which is in SS has ranking a multiple of 1515?

- [ ] {(3, 0, 0), (0, 0, 1)}{(3, 0, 0), (0, 0, 1)}
- [ ] {(3, 0, 1)}{(3, 0, 1)}
- [ ] {(3, 0, 1), (0, 0, 1)}{(3, 0, 1), (0, 0, 1)}
- [ ] {(0, 0, 1)}{(0, 0, 1)}
- [ ] {(9, 25, 0), (0, 0, 1)}{(9, 25, 0), (0, 0, 1)}
- [ ] {(25, 9, 0), (0, 0, 1)}{(25, 9, 0), (0, 0, 1)}
- [ ] {(9, 0, 0), (0, 25, 0), (0, 0, 1)}{(9, 0, 0), (0, 25, 0), (0, 0, 1)}
- [ ] {(25, 0, 0), (0, 9, 0), (0, 0, 1)}{(25, 0, 0), (0, 9, 0), (0, 0, 1)}

No, the answer is incorrect.

Score: 0

Accepted Answers:

{(3, 0, 0), (0, 0, 1)}{(3, 0, 0), (0, 0, 1)}

{(3, 0, 1)}{(3, 0, 1)}

{(3, 0, 1), (0, 0, 1)}{(3, 0, 1), (0, 0, 1)}

{(0, 0, 1)}{(0, 0, 1)}

{(9, 25, 0), (0, 0, 1)}{(9, 25, 0), (0, 0, 1)}

{(9, 0, 0), (0, 25, 0), (0, 0, 1)}{(9, 0, 0), (0, 25, 0), (0, 0, 1)}

Suppose that for a YouTube video there are 5 (in thousands) shares. What must be the number of comments (in thousands) so that the ranking of the video will be 4? [Note: Suppose your answer is 6000, then you have to enter 6 as the answer.]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

Check Answers